



UNIVERSITÄT
LEIPZIG

Natural Language Processing

Week 1: Introduction and Organisation

Leonie Weissweiler

09.04.2026

About Me

- Assistant Professor (Juniorprofessorin) for Natural Language Processing
- Can be emailed in either German or English
- No fixed office hours, please email
- New here, still figuring out administrative issues

Basic Information: Lecture

Lecture

- Where: Felix-Klein-Hörsaal
- When: Thursdays 9-11 c.t.
- Who: Leonie Weißweiler
- Moodle: only Vorlesung will be used, ignore the Übung pages
- Slides will be uploaded to moodle after each lecture

Basic Information: Tutorials

Tutorials

- Where: SG 3-12
- When: Wednesdays, either 11-13 c.t. or 15-17 c.t.
- Who: Toshiki Nakai, Jacob Suchardt
- Exercise sheet will be uploaded to moodle after each lecture
- Exercise sheet is voluntary and will be discussed after 6 days in the tutorials
- Questions will be exam-style but may eventually also include programming

Course Language

- English
- If you must, you can email me in German but email the TAs in English

Course Prerequisites

- Basic familiarity with python is useful for following along with programming exercises

(Preliminary) Semester Plan

- 09.04. Intro and NLP problems
- 16.04. Text processing
- 23.04. Corpora and tokenisation
- 30.04. N-gram language models
- 07.05. Transformers 1
- 21.05. Transformers 2
- 28.05. LLMs 1
- 04.06. LLMs 2
- 11.06. Morphology
- 18.06. Syntax
- 25.06. Semantics
- 02.07. Pragmatics
- 09.07. Outlook: Current Issues

Exam

- Week after the semester ends, exact date tbd
- Written, 90 minutes
- Similar to exercise sheets
- Second exam (Nachklausur) near the beginning of the winter semester tbd



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Questions about organisation?



Today's Lecture: Introduction and NLP Problems

Language and Technology

- **Computational Linguistics (CL)** is the study of using computational models to understand and process human language.
- **Natural Language Processing (NLP)** is a field at the intersection of linguistics and artificial intelligence that focuses on enabling computers to understand, generate, and manipulate human language.

Computational Linguistics vs Natural Language Processing

- Science
- Which theory is correct?
- Engineering
- Which theory is useful?

12

Related Fields

- Linguistics
- Statistics
- Psychology
- Machine Learning and AI

Textbooks

- Eisenstein: Natural Language Processing. MIT Press 2019
<https://github.com/jacobeisenstein/gt-nlp-class/tree/master/notes>
- Jurafsky & Martin: Speech and Language Processing. Prentice Hall 2000, 2009, 2025. <https://web.stanford.edu/~jurafsky/slp3/>
- Goldberg: Neural Network Methods for Natural Language Processing. Morgan & Claypool Publishers 2017.

More Applied Textbooks

- Rao & McMahan: Natural Language Processing with PyTorch. O'Reilly 2019. <https://github.com/delip/PyTorchNLPBook>
- Tunstall, Werra & Wolf: Natural Language Processing with Transformers. O'Reilly 2022. <https://github.com/nlp-with-transformers>

Scientific Venues

- Association for Computational Linguistics (ACL)
- Empirical Methods in Natural Language Processing (EMNLP)
- Regional ACLs: EACL, NAACL, AACL
- Non-ACL conferences. COLING, LREC
- Journals: Transactions of the ACL (TACL), Computational Linguistics (CL)
- All papers can be found in the ACL anthology:
<https://aclanthology.org/>

Software Packages

- Main (maybe only) programming language is Python
- Important packages to keep in mind for later:
 - spaCy
 - Stanza
 - NLTK
 - Scikit-learn
 - Hugging Face

Goals of Natural Language Processing

- 1. Aid humans in writing.

Correcting mistakes, formulating and paraphrasing text, transcription.

- 2. Identify texts related to spoken or written requests.

Text information retrieval, semantic text similarity, question answering.

- 3. Make sense of texts without reading the originals.

Categorization, information extraction, summarization, translation.

- 4. Instruct and be advised by a computer.

Audio interfaces (e.g., dialog systems, robotics), learning and assessment.

- 5. Converse with computers as if they were human.

Turing test, Conversational AI and chatbots, argumentation, affective computing.

- 6. Understanding the nature of language and its relation to (artificial) intelligence.

Pragmatics, causality, alignment and grounding, multimodality, simulation, linguistics.



Examples of NLP Systems

Writing: Spelling and Grammar Checking

Alan Turing

“Alan Mathison Turing (23 June 1912 – 7 June 1954) was an english mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Turing was highly influential in the developing of theoretical computer science, providing a formalisation of the concepts of algorithm and computatoin with the Turing machine, who can be considered a model of general-purpose computer. Turing is widely considered to been the farther of theoretical computer science and artificial intelligence. Despite these accomplishment he was ever fully recognised in his home country during his lifetime due to his homosexuality and because many of his work was covered by the Official Secrets Act.”

Can you spot any errors?

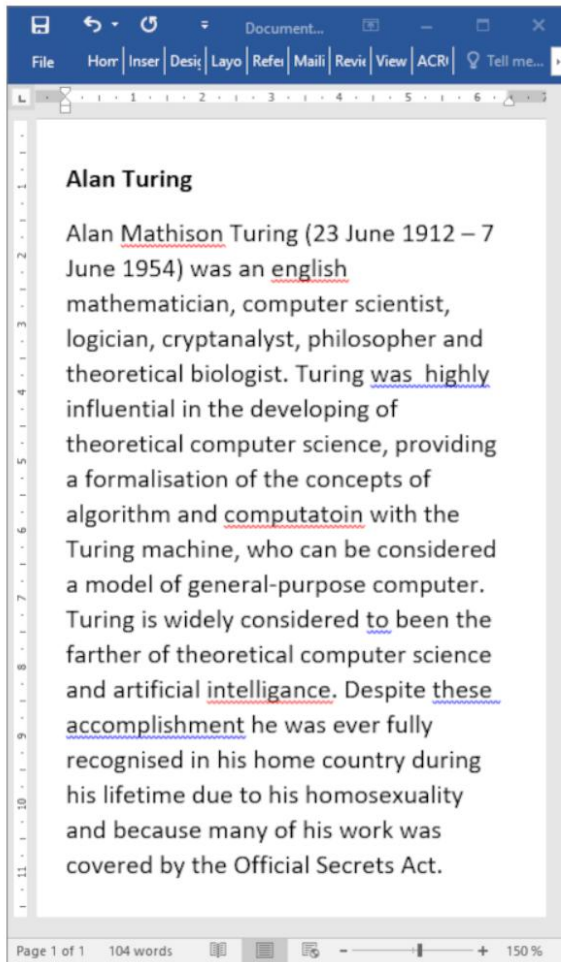
Writing: Spelling and Grammar Checking

Alan Turing

“Alan Mathison Turing (23 June 1912 – 7 June 1954) was an **english** mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Turing was highly influential in the **developing** of theoretical computer science, providing a formalisation of the concepts of algorithm and **computatoin** with the Turing machine, who can be considered a model **of general-purpose computer**. Turing is widely considered **to been** the **farther** of theoretical computer science and artificial **intelligence**. Despite **these accomplishment** he was **ever** fully recognised in his home country during his lifetime due to his homosexuality and because **many of his work** was covered by the Official Secrets Act.”

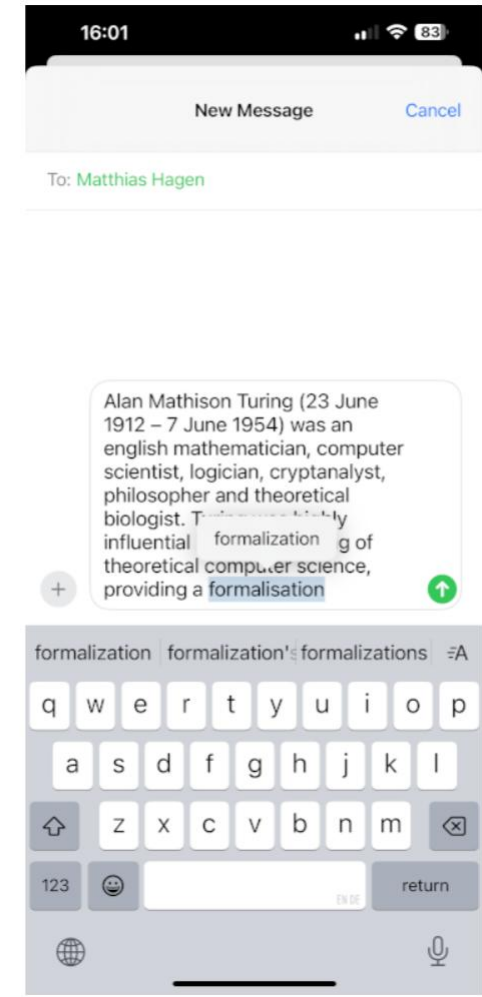
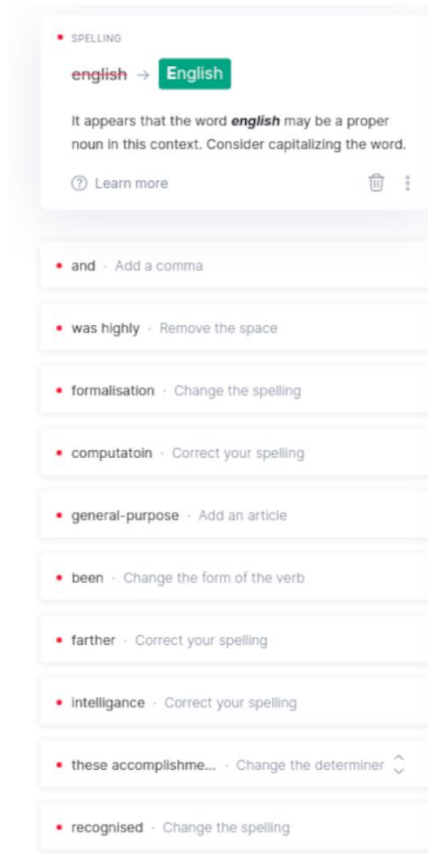
Can you spot any errors?

Writing: Spelling and Grammar Checking



Alan Turing

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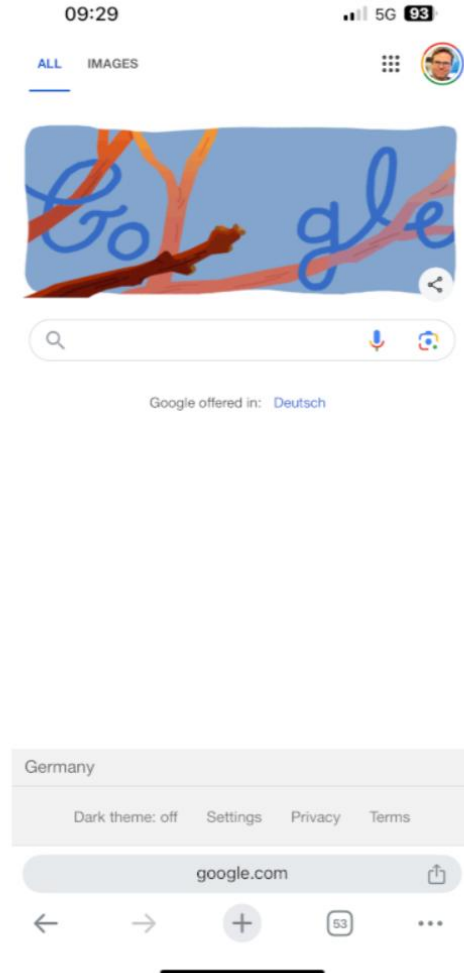
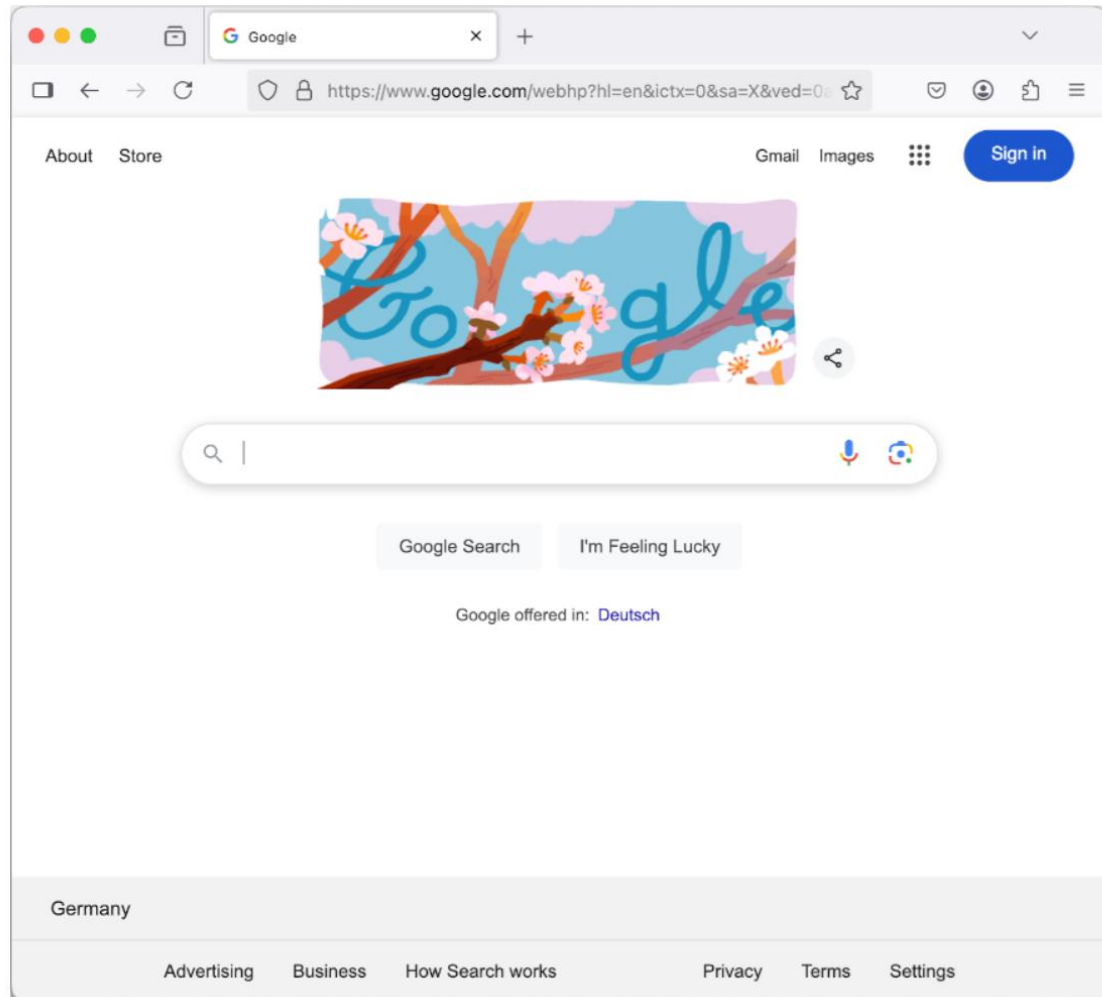
Remarks

- The text has been derived from a revision of the opening paragraph of the Alan Turing article on Wikipedia.
- Spelling and grammar checking includes detecting mistakes, highlighting them in the editor, and suggesting corrections. It enables the writer to review their own mistakes, correct them manually, and learn from them.
- On mobile devices, writing is supported by autocompleting the current word, the next words, and optionally correcting typing mistakes automatically. This can lead to erroneous corrections.
- Detected errors:
 - **english** should be capitalized (Word, Grammarly)
 - **and** could be preceded by an Oxford comma (Grammarly)
 - **was highly** should only have one space (Word, Grammarly)
 - **formalisation** could be switched to American English spelling (all, dependent on locale)
 - **computatoin** should be 'computation' (all)
 - **general-purpose** should be preceded by the article 'a' (Grammarly)
 - **to been** should be 'be' (Word, Grammarly; but Word for the wrong reason)
 - **farther** should be 'father' (Grammarly)
 - **intelligance** should be 'intelligence' (all)
 - **these accomplishment** should be 'these accomplishments' (Word, Grammarly)
 - **recognised** could be switched to American English spelling (all, dependent on locale)

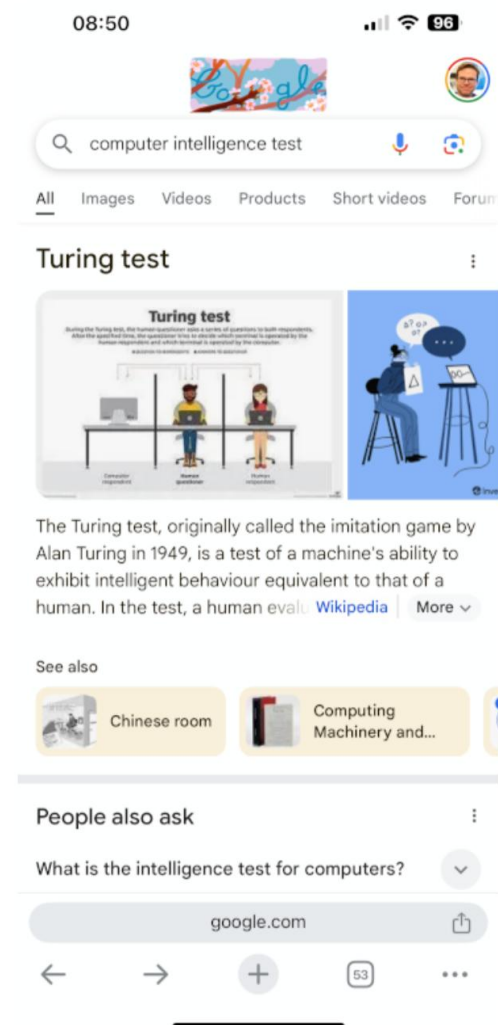
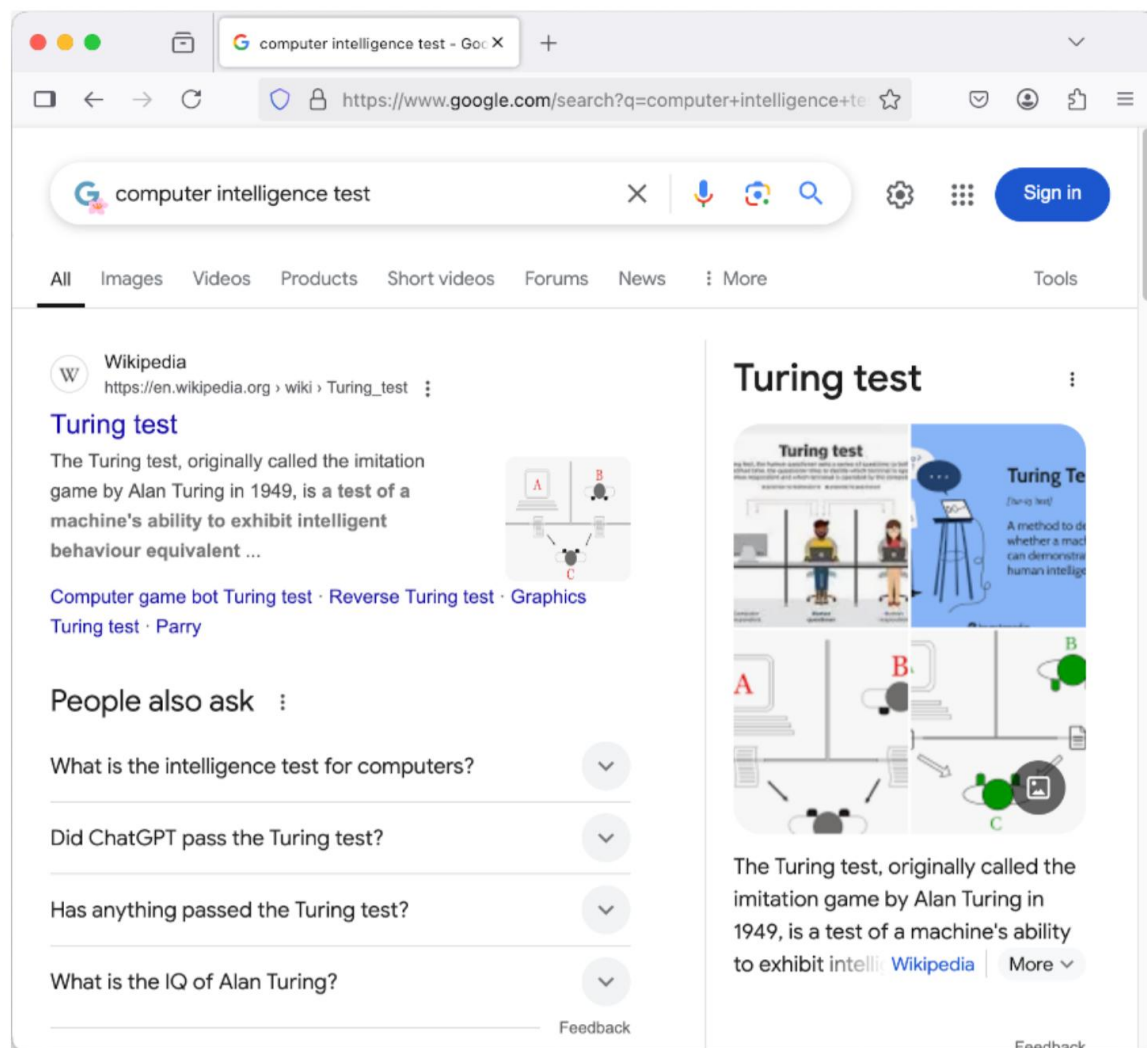
False detections/auto-completions and undetected errors

- Mathison is correctly spelled; false positive (Word, meanwhile fixed)
- developing should be 'development'; false negative (all)
- who should be 'which'; false negative (all)
- farther should not be 'farting'; false auto-completion (iPhone)
- ever should be 'never'; false negative (all)
- many should be 'much'; false negative (all)

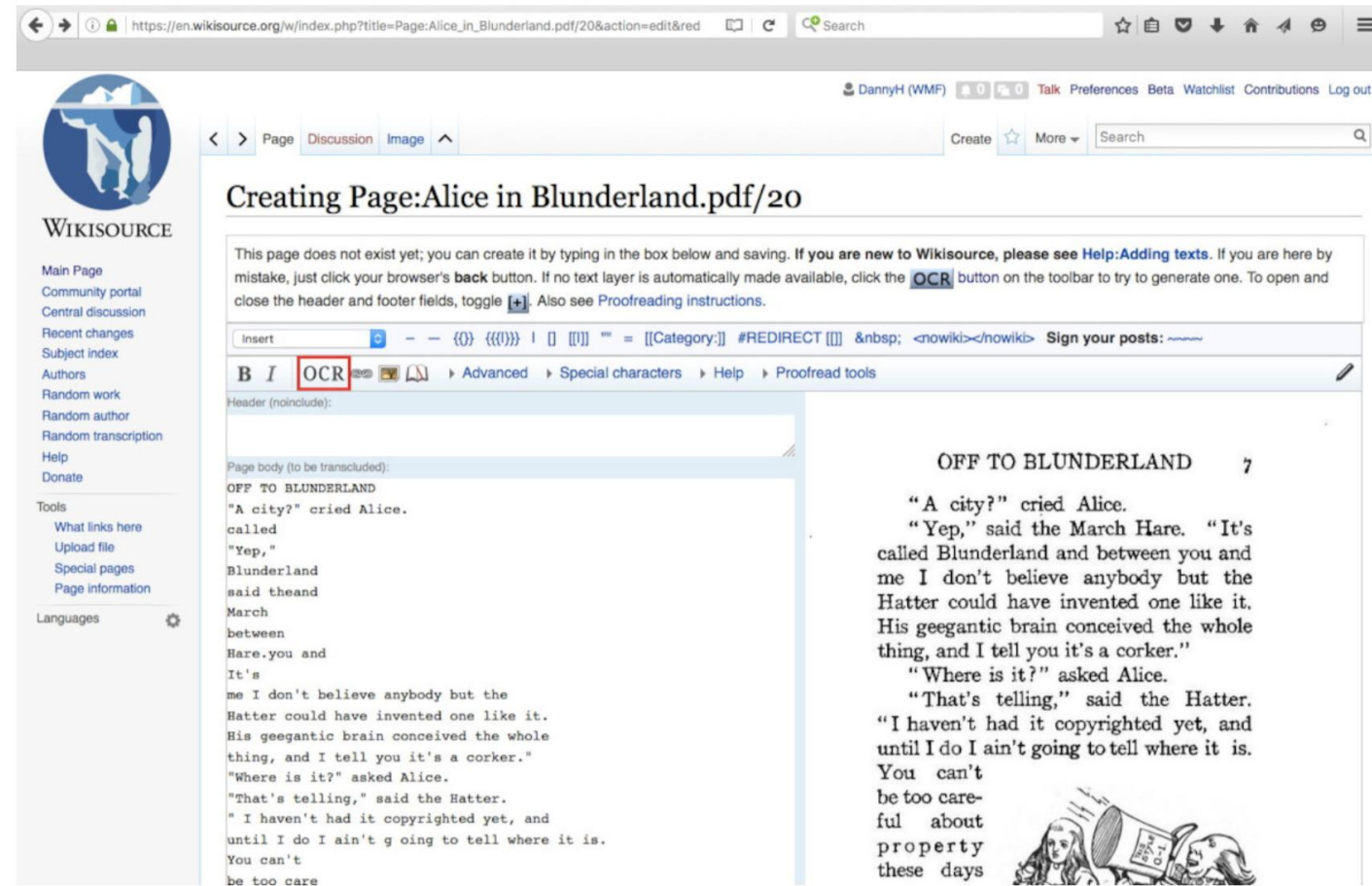
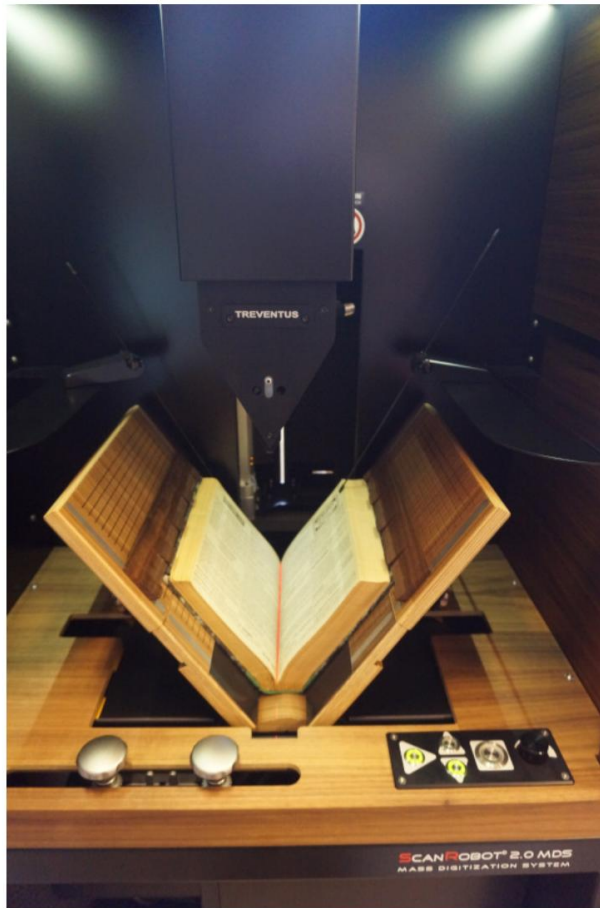
Search



Search



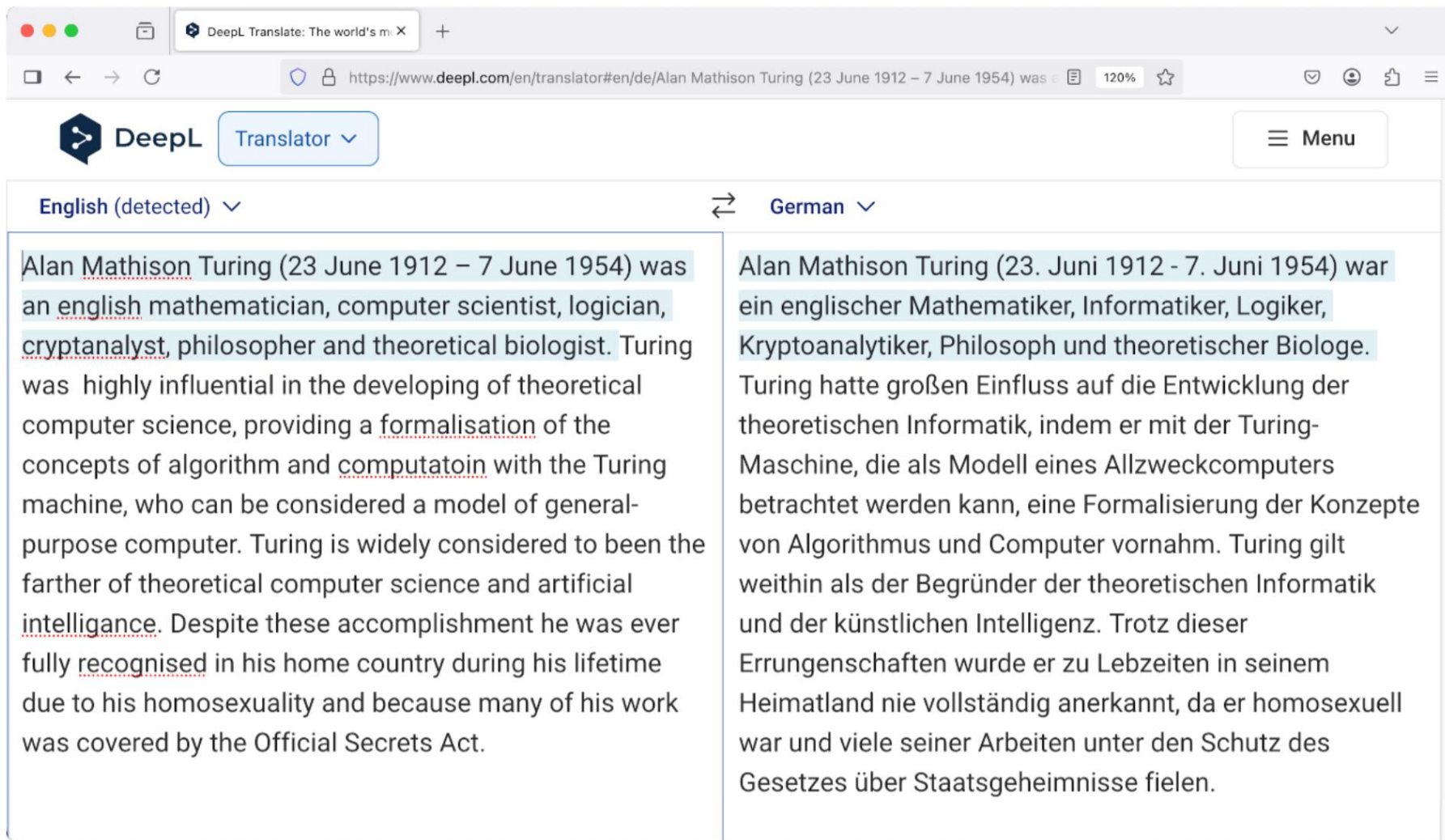
Optical Character Recognition



Optical Character Recognition



Machine Translation

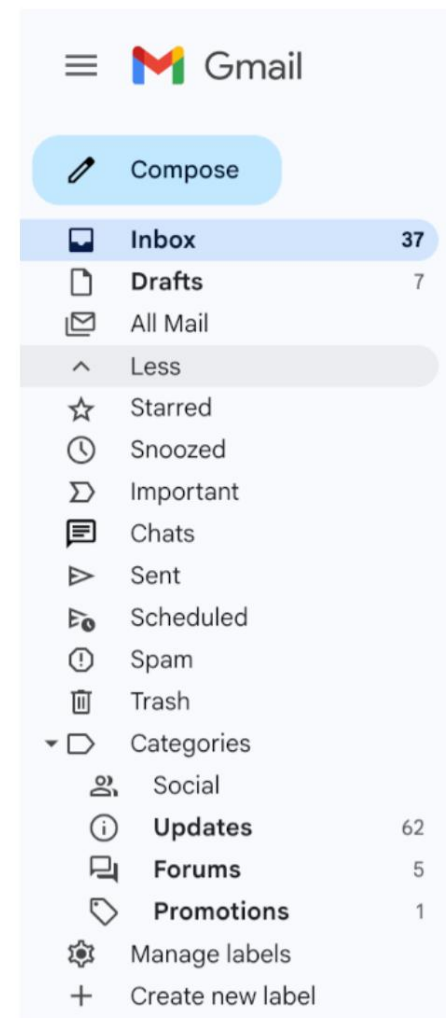
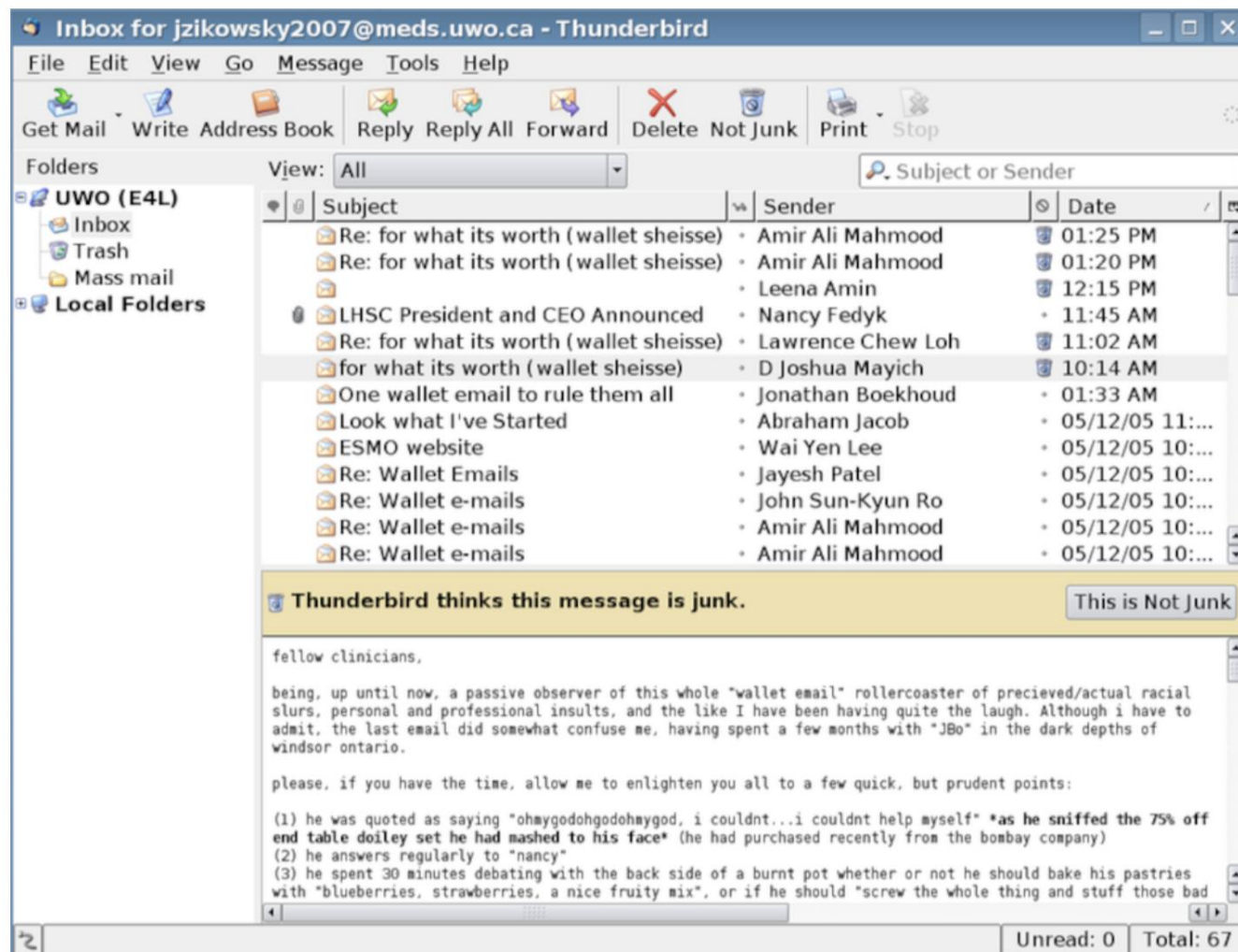


The screenshot shows the DeepL Translator web interface. The browser address bar displays the URL: [https://www.deepl.com/en/translator#en/de/Alan Mathison Turing \(23 June 1912 – 7 June 1954\) was](https://www.deepl.com/en/translator#en/de/Alan Mathison Turing (23 June 1912 – 7 June 1954) was). The page title is "DeepL Translate: The world's m...". The interface features a "Translator" button and a "Menu" button. The source language is "English (detected)" and the target language is "German". The input text on the left is: "Alan Mathison Turing (23 June 1912 – 7 June 1954) was an english mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Turing was highly influential in the developing of theoretical computer science, providing a formalisation of the concepts of algorithm and computatoin with the Turing machine, who can be considered a model of general-purpose computer. Turing is widely considered to been the farther of theoretical computer science and artificial intelligence. Despite these accomplishment he was ever fully recognised in his home country during his lifetime due to his homosexuality and because many of his work was covered by the Official Secrets Act." The output text on the right is the German translation: "Alan Mathison Turing (23. Juni 1912 - 7. Juni 1954) war ein englischer Mathematiker, Informatiker, Logiker, Kryptoanalytiker, Philosoph und theoretischer Biologe. Turing hatte großen Einfluss auf die Entwicklung der theoretischen Informatik, indem er mit der Turing-Maschine, die als Modell eines Allzweckcomputers betrachtet werden kann, eine Formalisierung der Konzepte von Algorithmus und Computer vornahm. Turing gilt weithin als der Begründer der theoretischen Informatik und der künstlichen Intelligenz. Trotz dieser Errungenschaften wurde er zu Lebzeiten in seinem Heimatland nie vollständig anerkannt, da er homosexuell war und viele seiner Arbeiten unter den Schutz des Gesetzes über Staatsgeheimnisse fielen."

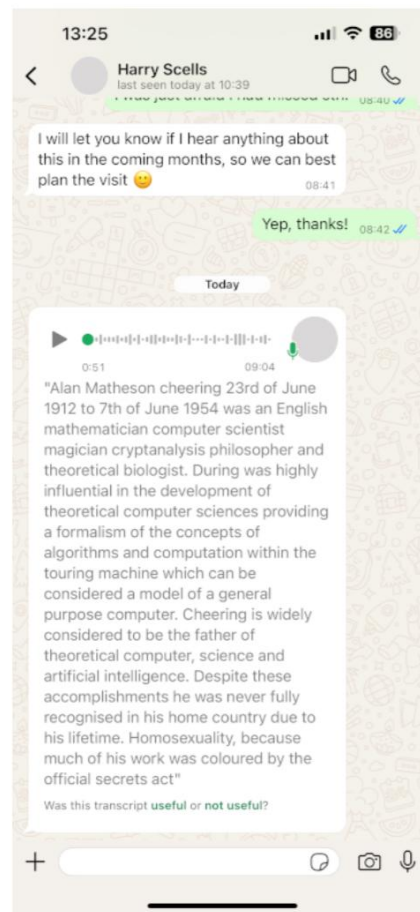
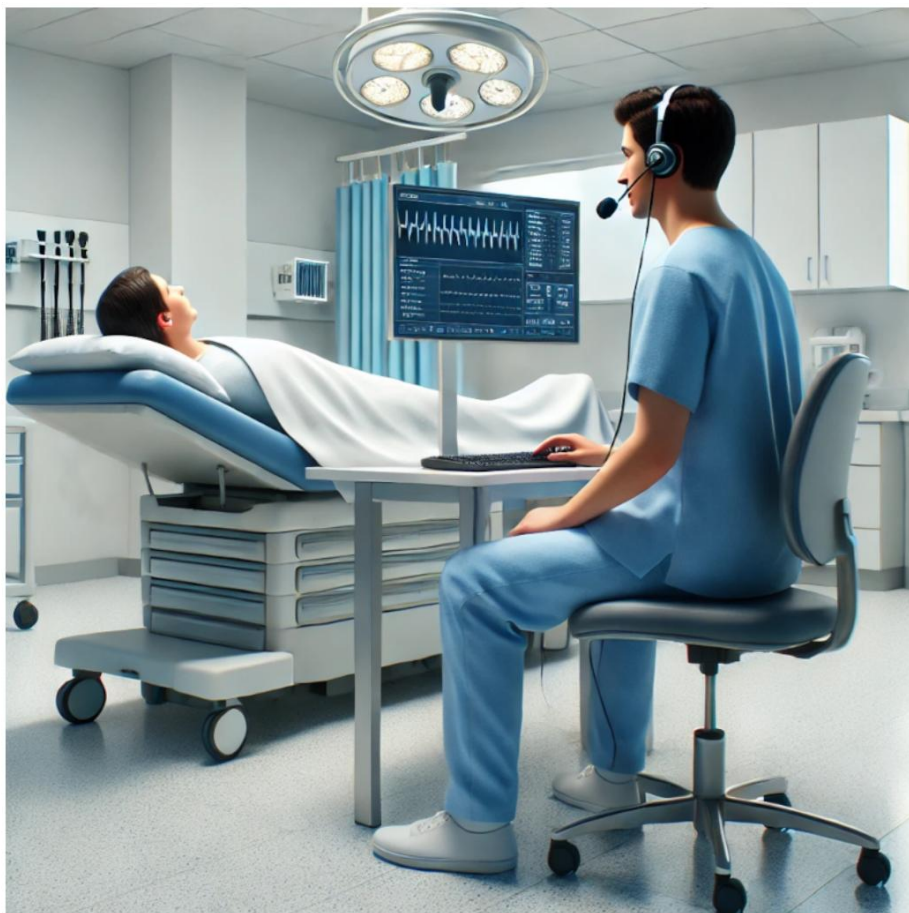
Machine Translation



Spam Detection and Email Categorisation



Transcription and Virtual Assistants



Business Use Cases

- Customer service and customer relation management

E.g., classification of customer support messages, dialogue systems and chatbots on web pages, or template-driven response generation.

- Business analytics and decision support

E.g., sentiment analysis of company- or product-related news or social media, summarization of user feedback, market forecast analysis for investments.

- Legal compliance and moderation of user-generated content

E.g., contract checking, invoice processing, classifying posts wrt. inappropriate content such as sexual messages or hate speech.

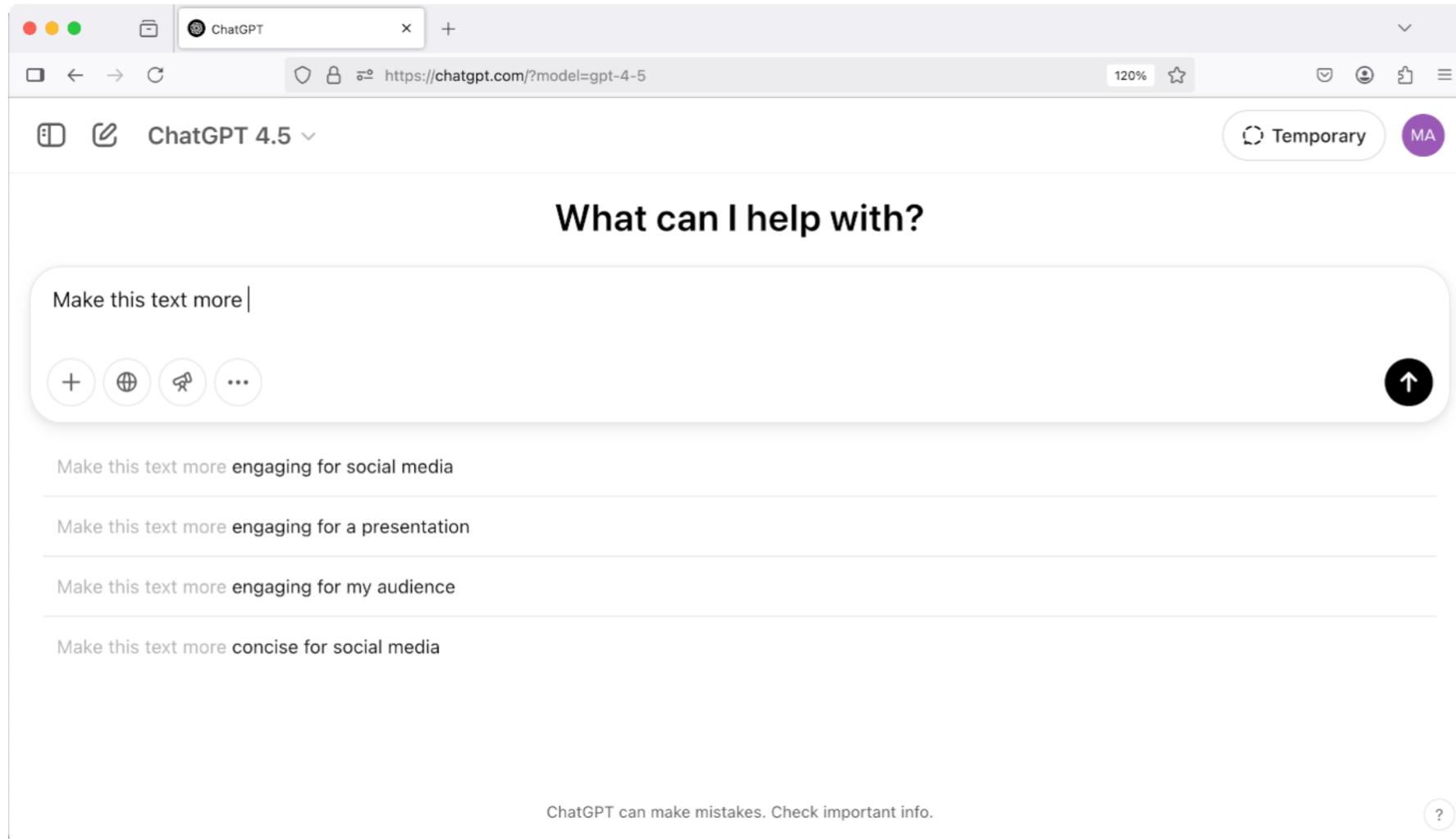
- Data journalism (e.g., on leaked documents)

E.g., search, information extraction, summarization.

Governance Use Cases

- Public health monitoring
E.g., social media monitoring for flu prediction
- Incident monitoring and disaster management
E.g., social media monitoring for damage assessment
- Criminal investigation and counter terrorism
E.g., style-based authorship analysis; search, classification, and summarisation of the dark web.
- Intelligence and espionage
E.g., monitoring and transcription of communication worldwide.

Chatbots



Chatbots

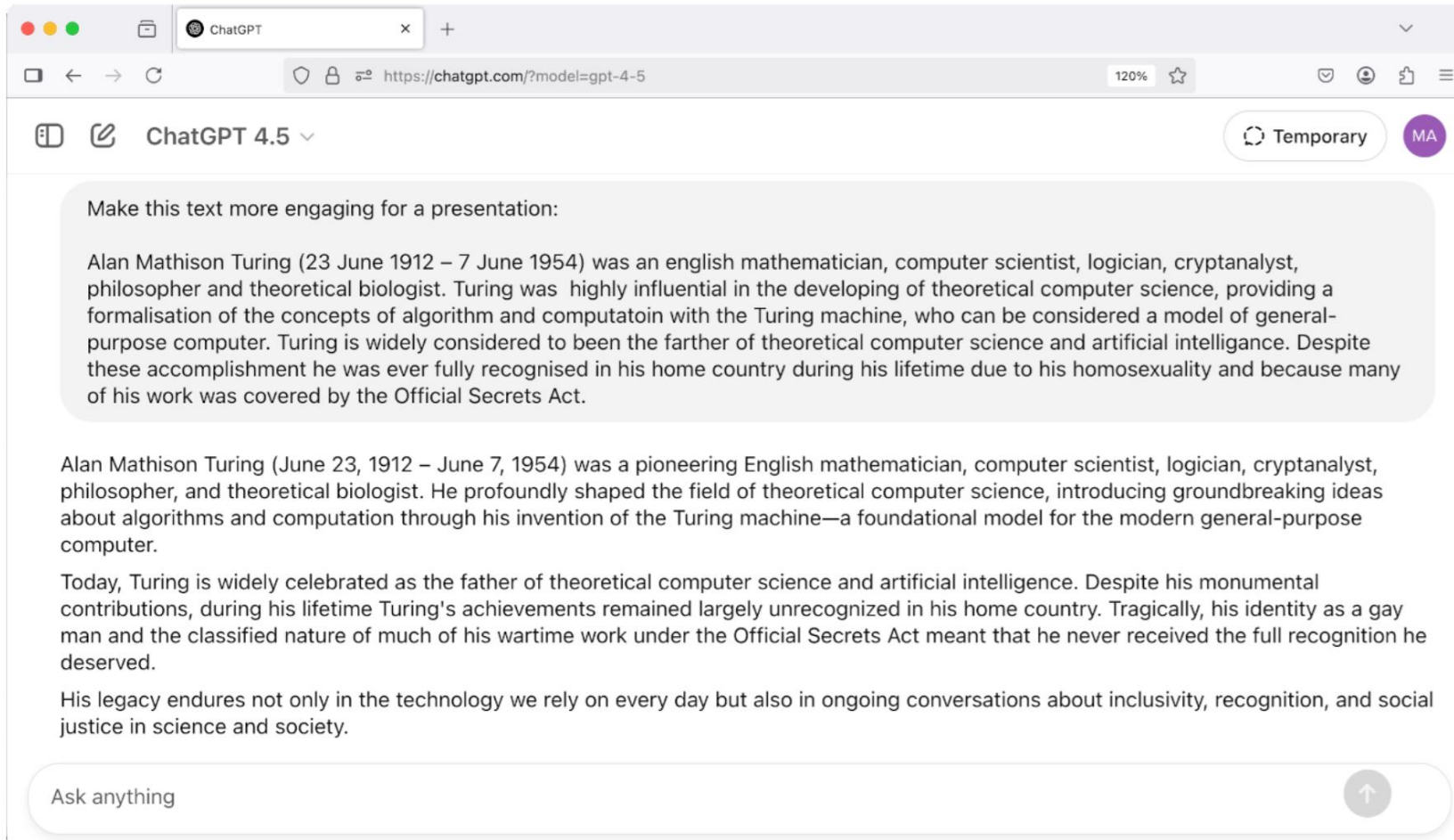
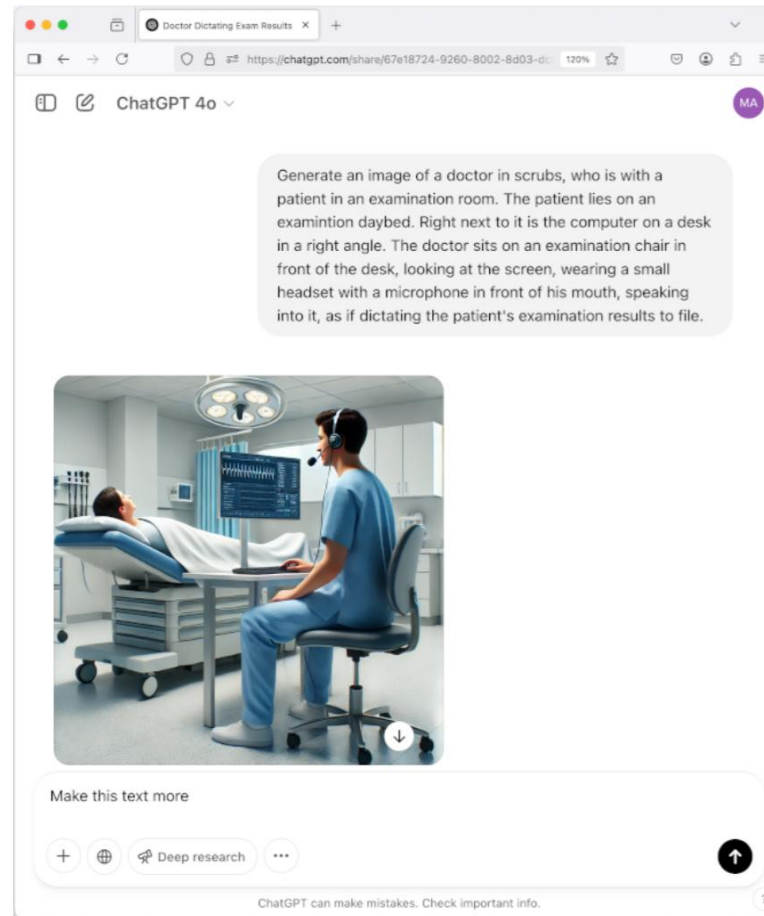
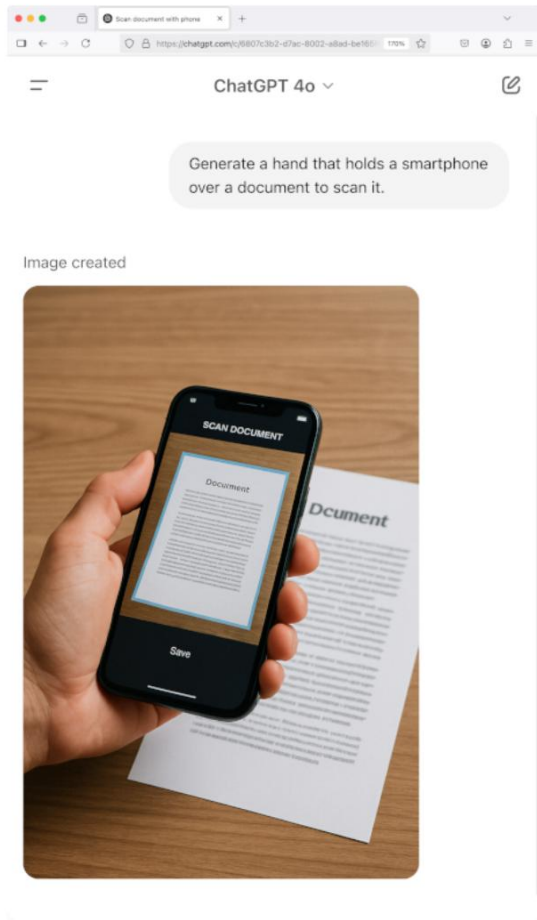


Image Generation from Text





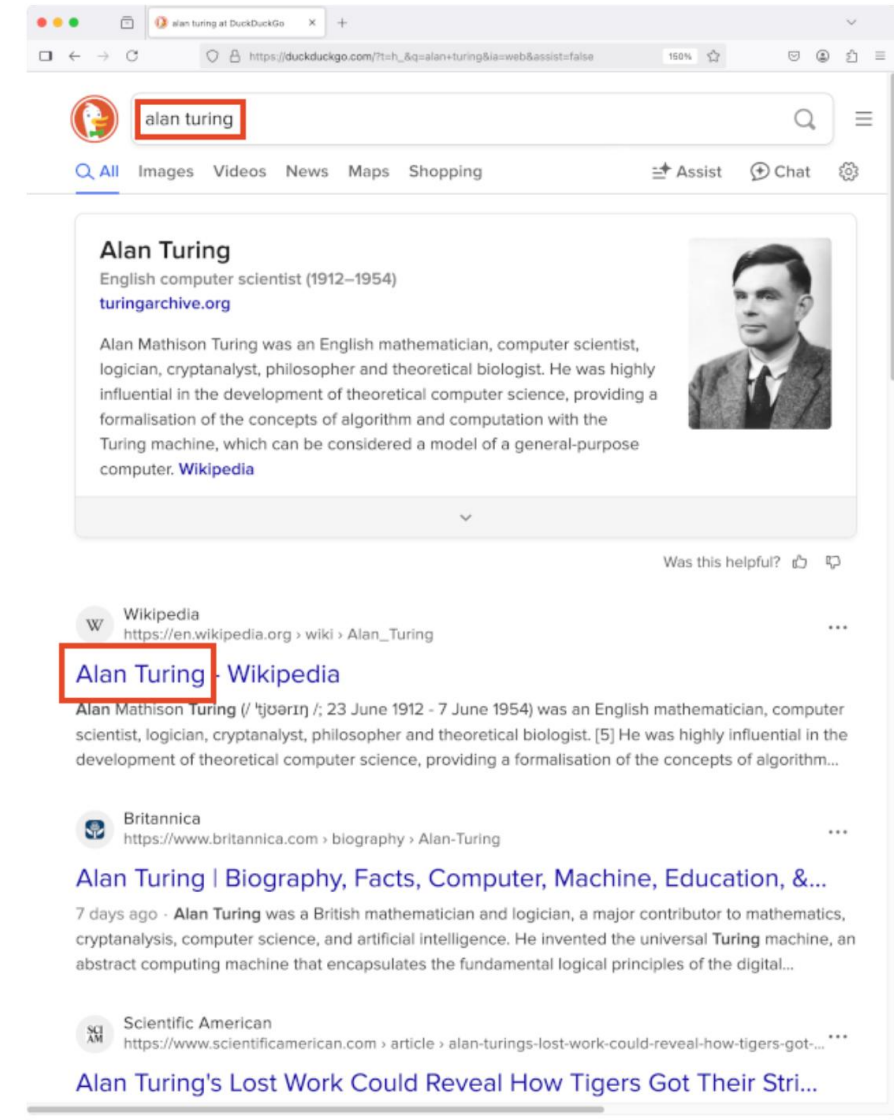
NLP Problems

Types of Tasks

- Analysis: the decoding of structured information from text
- Synthesis: the encoding of (structured) information into text

NLP Inside a Search Engine

- Search engines face the vocabulary mismatch problem between users and document authors.
- Users query one way, authors write another.
- Users still expect to find all relevant documents, regardless of how they write.
- Search engines implement a pipeline of basic NLP tasks to address this problem:
 - 1. Tokenization
 - 2. Lemmatization
 - 3. Stemming



Token and Sentence Splitting

Alan Mathison Turing (23 June 1912 - 7 June 1954) was
an English mathematician. He was highly influential
in the development of theoretical computer science,
providing a formalisation of the concepts of algorithm
and computation with the Turing machine, which can be
considered a model of a general-purpose computer.

Token and Sentence Splitting

Given a text as a character sequence

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Token and Sentence Splitting

Given a text as a character sequence, split it into tokens.

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Token and Sentence Splitting

Given a text as a character sequence, split it into sentences.

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Remarks

- The goal of tokenization is typically to identify the words in a text.
- A word is defined as “an element or unit of speech [or] language”. “Any of the sequences of one or more sounds or morphemes (intuitively recognized by native speakers as) constituting the basic units of meaningful speech used in forming a sentence or utterance in a language (and in most writing systems normally separated by spaces); a lexical unit other than a phrase or affix; an item of vocabulary, a vocable.”[Oxford English Dictionary]
- In semiotics, a token is “a particular and individual sign, as opposed to the type of which it is an instance”. [Oxford English Dictionary] Non-words can also be tokens (e.g., punctuation, number, abbreviation, etc.).
- Tokens are distinguished from types; the so-called type–token distinction. Example: “A rose is a rose is a rose.” In one sense of ‘word’, this sentence contains three words, but in another sense of ‘word’, it contains eight words. In the first sense, the words are types, and in the second sense, they are tokens. In simplified terms, a token may refer to a particular instance of a word printed on a page, while a type refers to the abstract form it instantiates. [Wetzel 2006]
- Splitting a text into a sequence of tokens is typically the first step in solving a real-world problem like suggesting spelling corrections using natural language processing.

Stemming and Lemmatisation

Given a tokenised text, normalise every word token by replacing it with its lemma (CoreNLP lemmatiser)

Alan Mathison Turing (23 June 1912 - 7 June 1954) ^{be} was
^a an English mathematician. ^{he} He ^{be} was highly influential
in the development of theoretical computer science,
^{provide} providing a formalisation of the ^{concept} concepts of algorithm
and computation with the Turing machine, which can be
^{consider} considered a model of a general-purpose computer.

Stemming and Lemmatisation

Given a tokenised text, normalise every word token by replacing it with its stem (Porter stemmer)

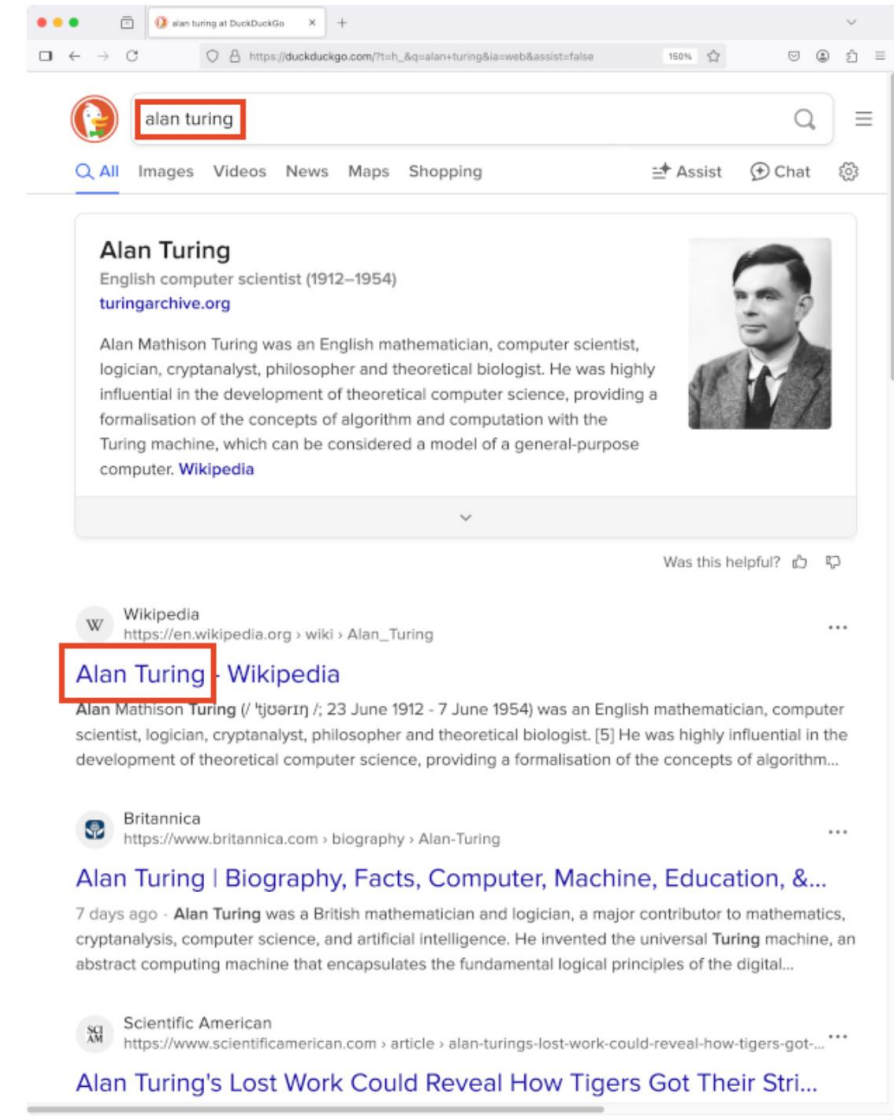
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Remarks

- Lemmatisation is the task of reducing a word token to its lemma, “a lexical item as it is presented, usually in a standardized form, in a dictionary entry; a definiendum”. [Oxford English Dictionary]
- Stemming is the task of reducing a word token to its stem, “that part of an inflected word that remains unchanged in the process of inflection”. [Oxford English Dictionary]
- Lemmatisation exchanges irregular inflections, while stemming only removes affixes from a word token.
- Stemming is typically based on a set of rules serving as heuristics to remove inflections from word tokens. These rules may fail to map word tokens with a common stem to that stem, and they may also map word tokens with lexicographically different stems to the same stem candidate.
- The goals of lemmatization and stemming are to ease semantic matching in downstream steps when solving a real-world problem.

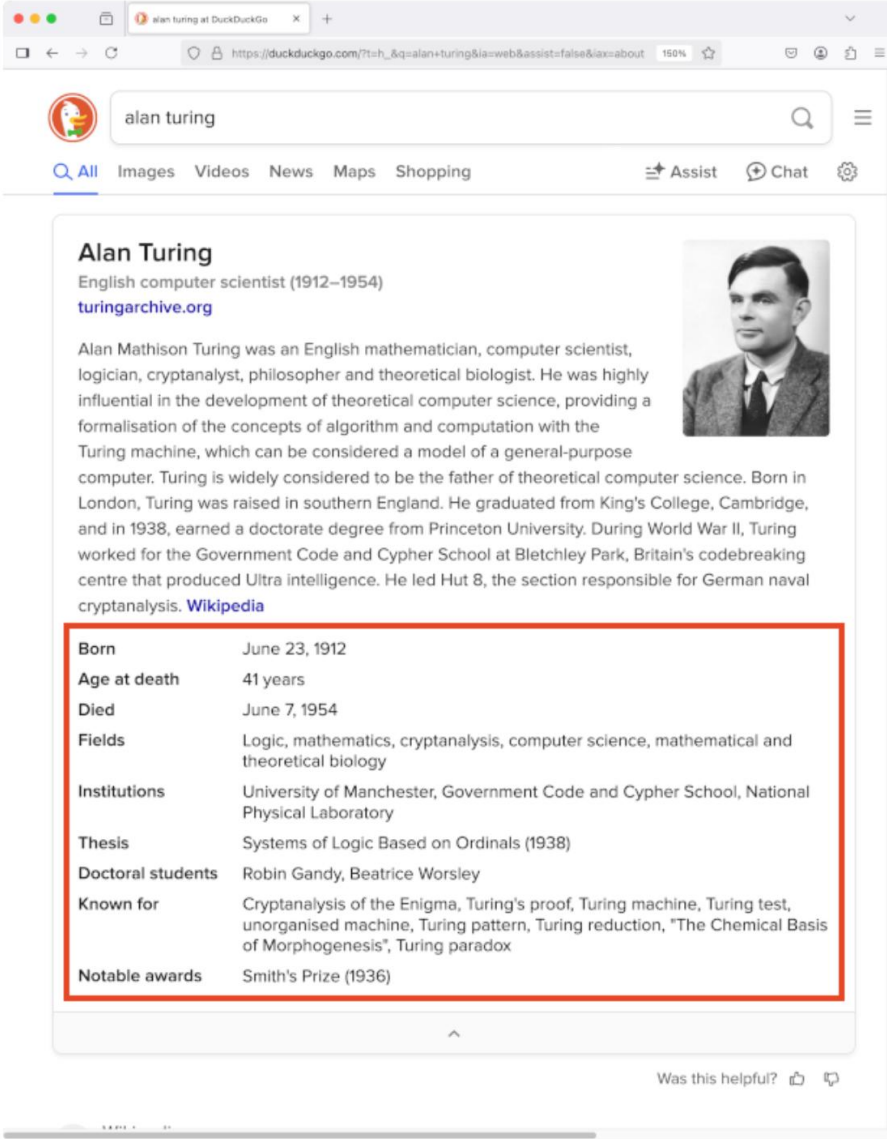
NLP Inside a Search Engine

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 - 2. Lemmatization
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NLP Inside a Search Engine

- Search engines return “knowledge panels” along search results.
- This relieves users from browsing the search results for basic facts.
- Unless it is collected manually, knowledge can be extracted from text using this pipeline:
 - 1. Tokenization
 - 2. POS tagging
 - 3. Named entity recognition
 - 4. Reference resolution
 - 5. Entity relation extraction



The screenshot shows a DuckDuckGo search result for "alan turing". The search bar at the top contains "alan turing". Below the search bar, there are tabs for "All", "Images", "Videos", "News", "Maps", and "Shopping". To the right of these tabs are links for "Assist", "Chat", and a settings icon. The main content area features a knowledge panel for Alan Turing. It includes a portrait of Alan Turing, his name "Alan Turing", and his title "English computer scientist (1912–1954)" with a link to "turingarchive.org". A paragraph of text describes his life and work, mentioning his role in the development of the Turing machine and his work at Bletchley Park. Below this text is a table of key facts about Alan Turing, which is highlighted with a red border. The table lists his birth date, age at death, death date, fields of study, institutions, thesis, doctoral students, known for, and notable awards. At the bottom right of the knowledge panel, there is a link to "Wikipedia".

Born	June 23, 1912
Age at death	41 years
Died	June 7, 1954
Fields	Logic, mathematics, cryptanalysis, computer science, mathematical and theoretical biology
Institutions	University of Manchester, Government Code and Cypher School, National Physical Laboratory
Thesis	Systems of Logic Based on Ordinals (1938)
Doctoral students	Robin Gandy, Beatrice Worsley
Known for	Cryptanalysis of the Enigma, Turing's proof, Turing machine, Turing test, unorganised machine, Turing pattern, Turing reduction, "The Chemical Basis of Morphogenesis", Turing paradox
Notable awards	Smith's Prize (1936)

Part-of-Speech Tagging

Given a tokenised text, identify the part of speech of each token.

NP	NP	NP	-LRB-	CD	NP	CD	HYPH	CD	NP	CD	-RRB-	VBD
Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954	)	was
DT	NP	NN	.	PRP	VBD	RB	JJ					
an	English	mathematician	.	He	was	highly	influential					
IN	DT	NN	IN	JJ	NN	NN	.					
in	the	development	of	theoretical	computer	science	.					

Tag	Description
-LRB-	Left round bracket
-RRB-	Right round bracket
.	Sentence terminator
CD	Cardinal number
DT	Determiner
HYPH	Hyphenation
IN	Preposition or subordinating conjunction

Tag	Description
JJ	adjective or numeral, ordinal
NN	noun, singular or mass
NP	noun, proper, singular
PRP	pronoun, personal
RB	Adverb
VBD	verb, past tense

Named Entity Recognition

Given a POS-tagged text, identify token sequences that name real-world objects.

PERSON				DATE				DATE					
Alan	Mathison	Turing		(	23	June	1912	-	7	June	1954)	was	¶
NATIONALITY		OCCUPATION											
an	English	mathematician.		He	was	highly	influential	¶					
				FIELD-OF-WORK									
in	the	development	of	theoretical	computer	science.							

Coreference Resolution

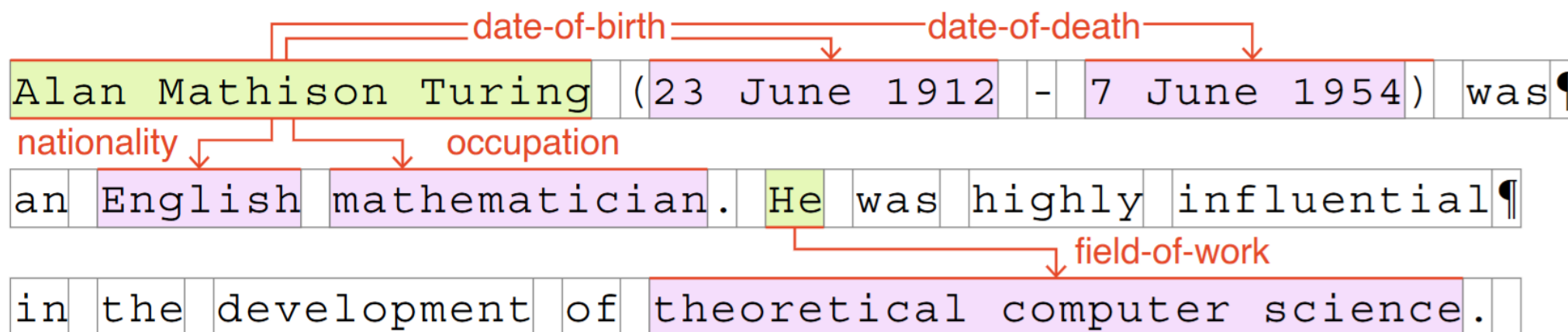
Given a POS-tagged text, map entities to their coreferents.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was
an English mathematician. He was highly influential
in the development of theoretical computer science.

A red arrow originates from the word 'He' in the second sentence and points back to the name 'Alan Mathison Turing' in the first sentence, indicating that they refer to the same entity.

Entity Relation Extraction

Given a NER-tagged text with coreferences, identify relations among entities.



The goal of entity relation extraction is to identify knowledge triples of the form (subject, predicate, object) which form the basis of modern knowledge bases.

Types of Tasks

- Analysis: the decoding of structured information from text
- Synthesis: the encoding of (structured) information into text

Autocomplete

- Typing on mobile devices is relatively slow.
- Many users are further slowed down by the use of logographic language and/or disabilities.
- A key input method includes predicting the currently typed word and/or the next word.
- Today, many systems construct a statistical language model for this purpose as follows:
 - 1. Corpus construction
 - 2. Tokenization
 - 3. Token k-gram counting
 - 4. Next token prediction



Token n-gram Counting

Given a tokenised text, tabulate its token n-gram counts, where a n-gram is a token sequence of length n .

Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954)	was	¶
an	English	mathematician.	He	was	highly	influential	¶					
in	the	development	of	theoretical	computer	science.						

Token n-gram Counting

Given a tokenised text, tabulate its token unigram counts, where a unigram is a token sequence of length 1.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was
an English mathematician. He was highly influential
in the development of theoretical computer science.

Token n-gram Counting

Given a tokenised text, tabulate its token bigram counts, where a bigram is a token sequence of length 2.

Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954)	was	¶
------	----------	--------	---	----	------	------	---	---	------	-------	-----	---

an	English	mathematician.	He	was	highly	influential	¶
----	---------	----------------	----	-----	--------	-------------	---

in	the	development	of	theoretical	computer	science.	
----	-----	-------------	----	-------------	----------	----------	--

Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954)	was	¶
------	----------	--------	---	----	------	------	---	---	------	-------	-----	---

an	English	mathematician.	He	was	highly	influential	¶
----	---------	----------------	----	-----	--------	-------------	---

in	the	development	of	theoretical	computer	science.	
----	-----	-------------	----	-------------	----------	----------	--

Token n-gram Counting

Given a tokenised text, tabulate its token trigram counts, where a trigram is a token sequence of length 3.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was

an English mathematician. He was highly influential

in the development of theoretical computer science.

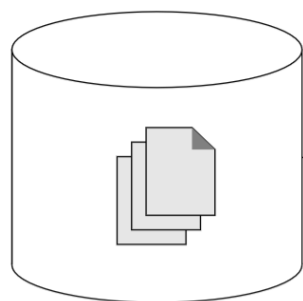
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an English mathematician. He was highly influential

in the development of theoretical computer science.

Token n-gram counting

Given a tokenised corpus, tabulate its token n-gram counts.



3-gram	Count
Alan Mathison Turing	2,800
Mathison Turing (	52
Turing (23	<42
⋮	⋮

How scalable is token n-gram counting?

Token n-gram counting

1-gram	Count	2-gram	Count	3-gram	Count
Alan	24,000,000	Alan Mathison	4,500	Alan Mathison Turing	2,800
Mathison	110,000	Mathison Turing	3,000	Mathison Turing (	52
Turing	1,300,000	Turing (	1,400	Turing (23	<42
(	3,300,000,000	(23	330,000	(23 June	883
23	320,000,000	23 June	670,000	23 June 1912	612
June	210,000,000	June 1912	51,000	June 1912 -	1,400
1912	12,000,000	1912 -	190,000	1912 - 7	92
-	19,000,000,000	- 7	870,000	- 7 June	737
7	760,000,000	7 June	550,000	7 June 1954	1,100
1954	20,000,000	June 1954	100,000	June 1954)	1,900
)	3,300,000,000	1954)	640,000	1954) was	109
was	3,700,000,000	) was	41,000,000	) was an	9,100
an	2,500,000,000	was an	41,000,000	was an English	170,000
English	270,000,000	an English	6,000,000	an English mathematician	7,600
mathematician	2,200,000	English mathematician	36,000	English mathematician .	1,100
.	39,000,000,000	mathematician .	130,000	mathematician . He	7,400
He	2,400,000,000	. He	320,000,000	. He was	47,000,000
highly	73,000,000	He was	270,000,000	He was highly	110,000
influential	9,300,000	was highly	1,600,000	was highly influential	12,000
in	15,000,000,000	highly influential	170,000	highly influential in	28,000
the	43,000,000,000	influential in	750,000	influential in the	220,000
development	420,000,000	in the	3,500,000,000	in the development	8,500,000
of	25,000,000,000	the development	57,000,000	the development of	44,000,000
theoretical	25,000,000	development of	78,000,000	development of theoretical	19,000
computer	240,000,000	of theoretical	1,500,000	of theoretical computer	30,000
science	260,000,000	theoretical computer	300,000	theoretical computer science	310,000
		computer science	12,000,000	computer science .	540,000
		science .	11,000,000		



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Questions?